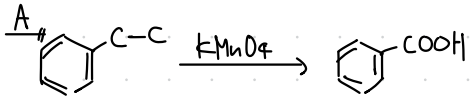
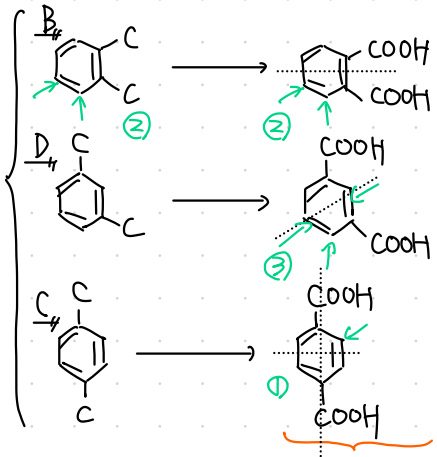
4-1　C_8H_{10}　$U=4$ ○×1

エチルベンゼン A 

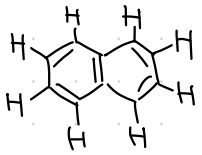
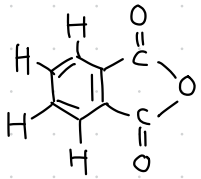
o, m, p-キシレン


◆ $C_8H_6O_3$（無水フタル酸）

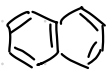
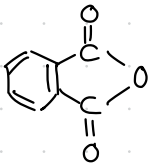
◆ $C_8H_6O_4$（フタル酸の異性体）

内2　(Ox)　$O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$　) ×9

(Red)　$C_{10}H_8 + 7H_2O \longrightarrow C_8H_4O_3 + 2CO_2 + 18H^+ + 18e^-$　) ×2

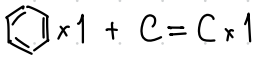



---

2  + $9O_2 \longrightarrow$ 2  + $4CO_2 + 4H_2O$ //

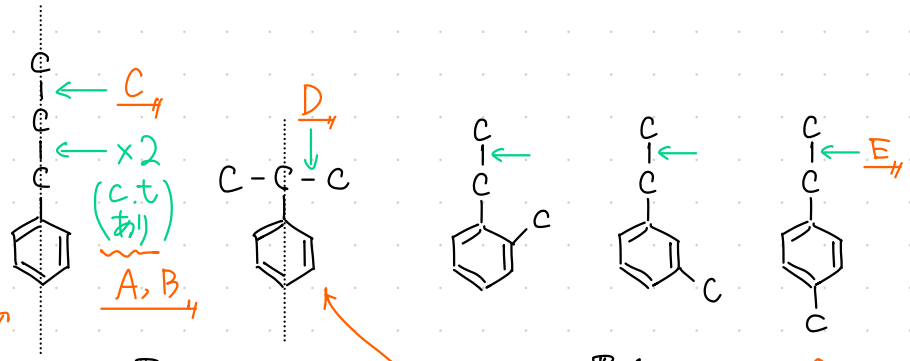
（酸化だから，余ったC $\xrightarrow{O_2}$ CO_2，
H $\xrightarrow{O_2}$ H_2O になると考えて，
係数合わせても良い）

4-(2)　C_9H_{10}　$U=5$

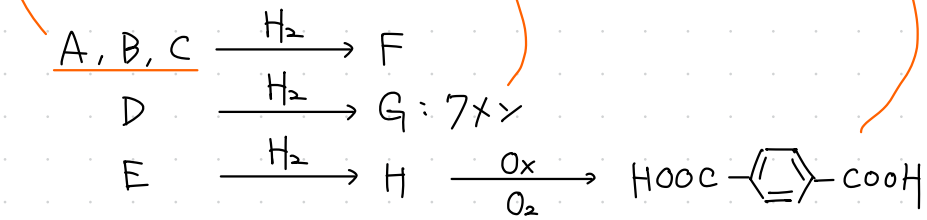


(U=4)　(U×1)

問2.



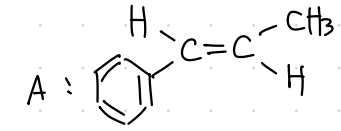
<1置換体>　<2置換体>

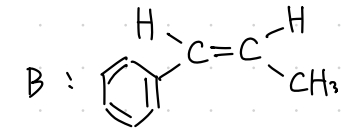


A, B：シス・トランス異性体

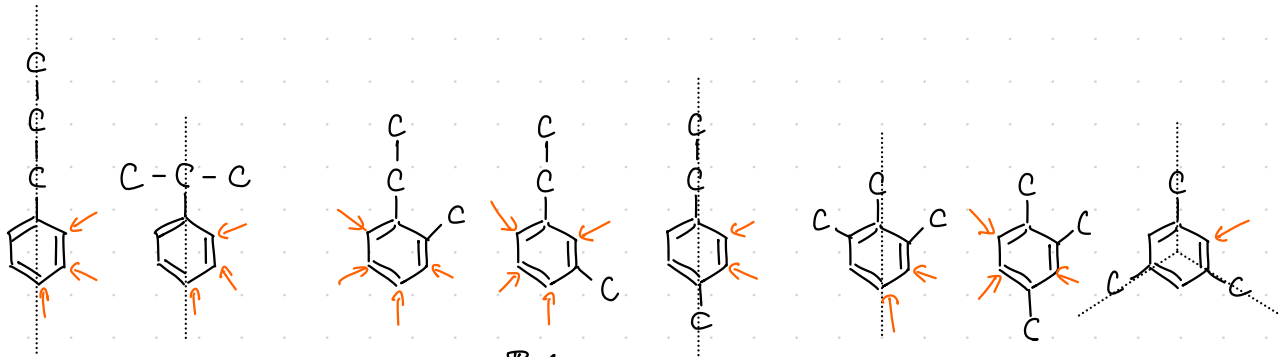
融点 A > B

トランス　シス





問1.　8　　問3.　22



<1置換体>
Cl置換：6

<2置換体>
Cl置換：10

<3置換体>
Cl置換：6

4-③

問1　$C_\ell H_m O_n + \underline{\quad\quad} O_2 \longrightarrow \ell\, CO_2 + \frac{m}{2} H_2O$

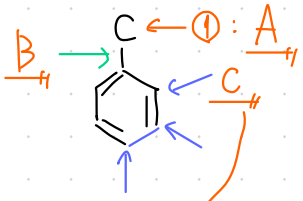
係数比　$\ell : \frac{m}{2} = 7 : 4$　　$\therefore (\ell, m) = (7, 8)$

$\therefore (C_7H_8O_n)_k = (92 + 16n)k = 108$

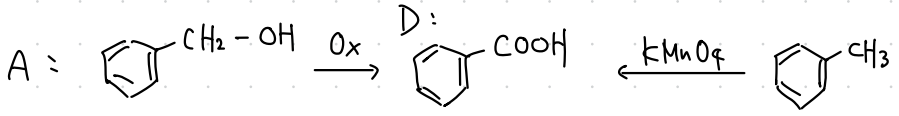
$\therefore (n, k) = (1, 1)$　　$\underline{C_7H_8O}$

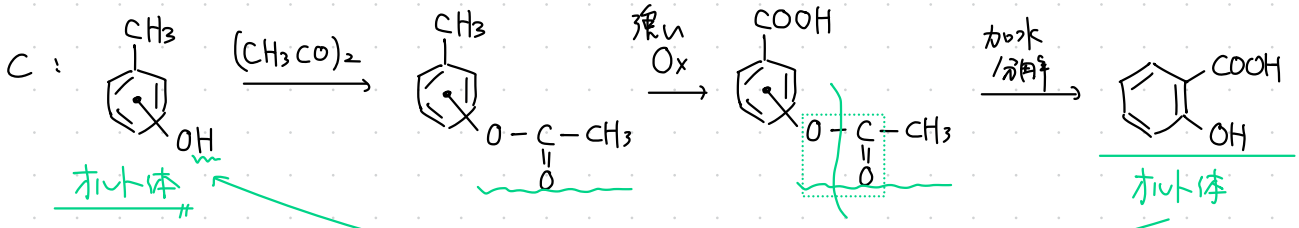
問2.　C_7H_8O　　$U = 4$　$\underline{\quad\quad}$ ×1

- アルコール
- フェノール類
- エーテル



	A	B（エーテル）	C
Na	H_2↑	×	
NaOH	×	とける	
	アルコール	フェノール類	

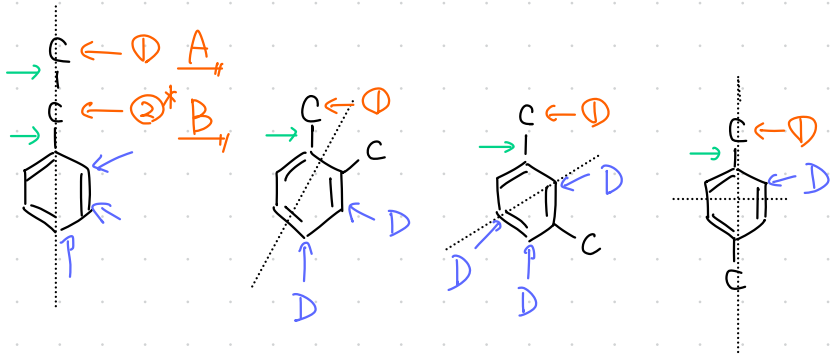




オルト体　　オルト体

4-④　$C_8H_{10}O$　U=4 (○ + 1)

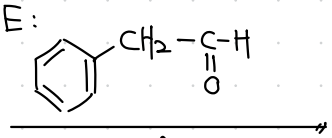
- アルコール
- フェノール類
- エーテル



第1級アルコール

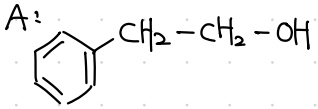
	A	B	C	D
置換基	1, 2			3

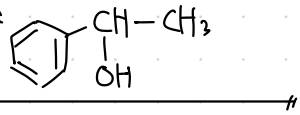
おだやか Ox　E アルデヒド

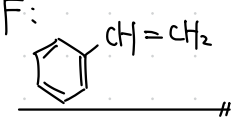
E:


↑ おだやか酸化

脱水　F　F　A と B は同一C骨格 :

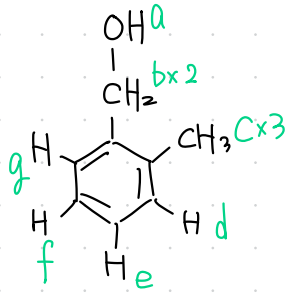
A:


B:


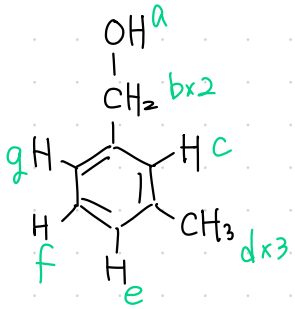
→ F:


$FeCl_3$ aq	−	+

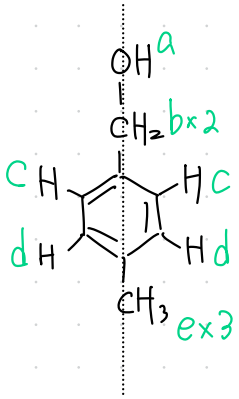
C : 1or2置換体のアルコール



7種 ×

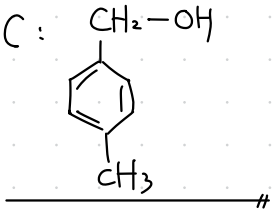


7種 ×



5種 (1:2:2:2:3)

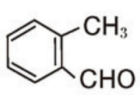
=

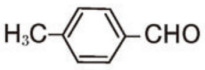
C:


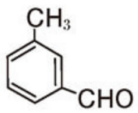
内2

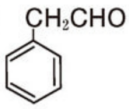
D: フェノール類 (3置換体), 6

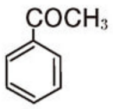
4－5

(1) A 

B 

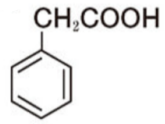
C 

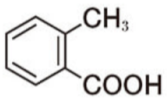
D 

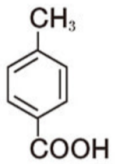
E 

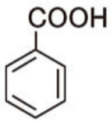
(2) Gはp-置換体であるため，o-，m-置換体よりも分子の形が対称的である。したがって，結晶化しやすいため，融点は最も高くなる。

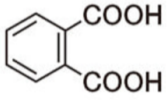
4－6

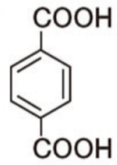
A 

B 

C 

D 

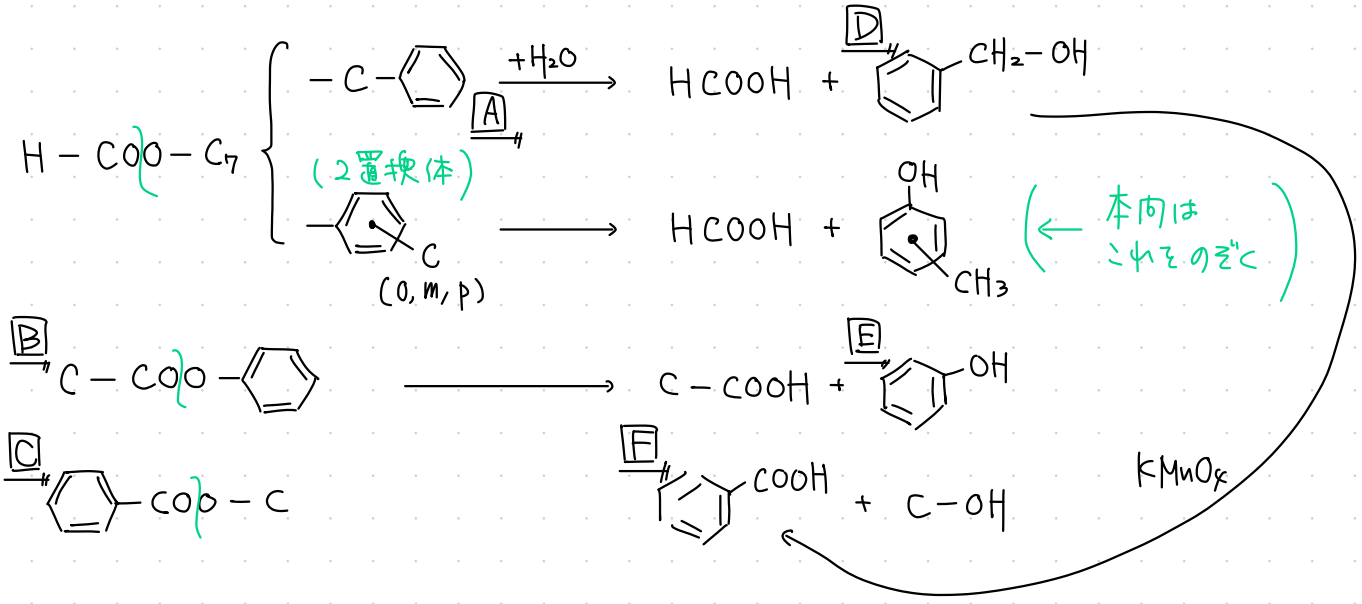
E 

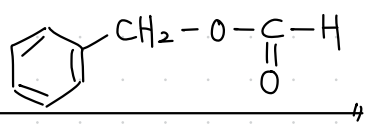
F 

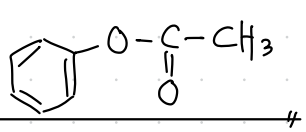
4-7

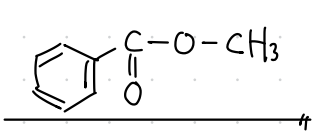
問1　$C_8H_8O_2$　$U=5$　$\bigcirc \times 1 + -\overset{\overset{O}{\|}}{C}-O- \times 1$

（$U=4$）　（$U=1$）

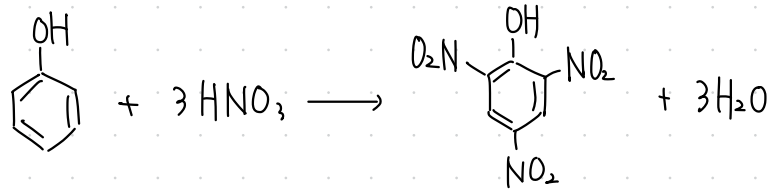


A： 

B： 

C： 

問2.　E

問3　 

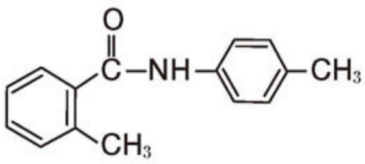
ピクリン酸

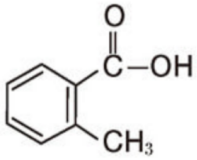
（2,4,6－トリニトロフェノール）

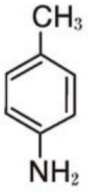
4 - 8

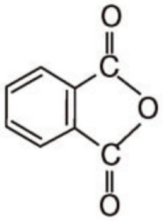
(1) $C_{15}H_{15}NO$　　$U = \frac{2\times15+2+1-15}{2} = 9$

(2)

A 

B 

C 

D 

$C_{15}H_{15}NO$　　$U = \frac{2\times15+2+1-15}{2} = 9 =$ アミド $-\overset{\overset{O}{\|}}{C}-NH-$ + ベンゼン環 ×2

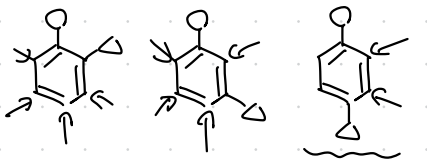
C_8以上

アミド A $\xrightarrow[H^+]{\text{加水分解}}$ ① カルボン酸 B（オルト体） $\xrightarrow{KMnO_4}$ B'（オルト体） $\xrightarrow[\triangle]{-H_2O}$ 酸無水物 D

$-\overset{\overset{O}{\|}}{C}-NH-$ $\xrightarrow{H^+}$ $-\overset{\overset{O}{\|}}{C}-OH$ + $-NH_3^+Cl^-$

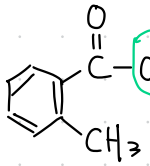
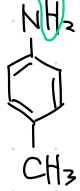
② アミン C の HCl 塩 $\xrightarrow[\text{Wittig}]{OH^-}$ アミン C

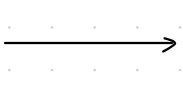
- 2置換体

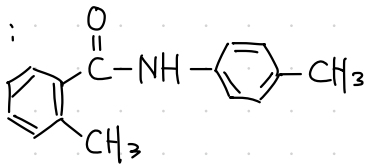


$-NH_2$, $-CH_3$ あり

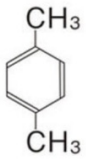
アミド A	カルボン酸 B　アミン C	
C_{15}	C_8以上	C_7以上
	↓	↓
	C_8	C_7

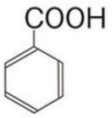
∴ B :  　C : 

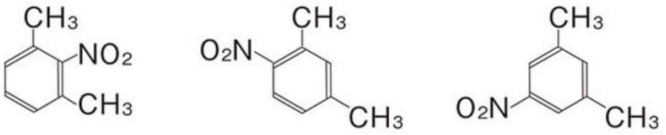


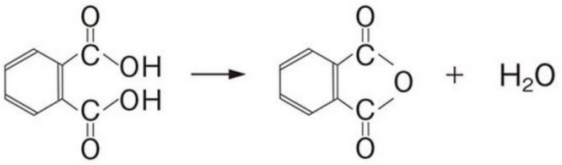
A : 

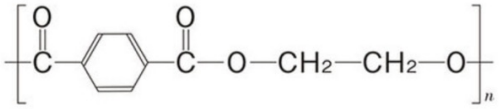
4－9

問1　化合物 D： 

問2　化合物 E： 

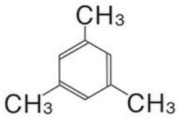
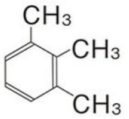
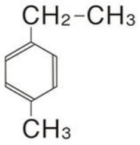
問3　 

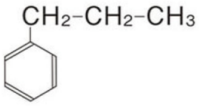
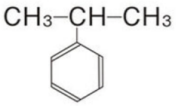
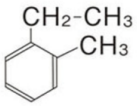
問4　 

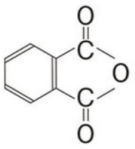
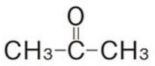
問5　 

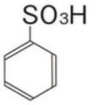
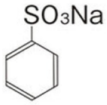
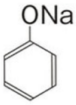
4－10

問1（ア） エチレングリコール（1,2エタンジオール）　（イ） 酸素
（ウ） ヨウ素

問2 A 
B 
C 

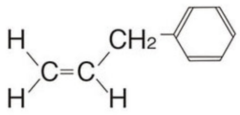
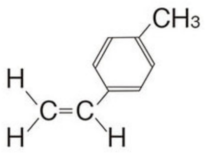
D 
E 
F 

K 
M 

問3 N 
O 
P 

4－11

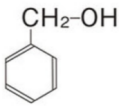
問1 2個　**問2** クメン

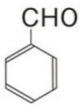
問3A 
E 

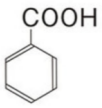
チ－12

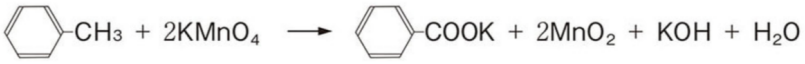
問1　気体…水素　　官能基…ヒドロキシ基

問2　試薬…$FeCl_3$　　色…紫

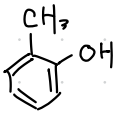
問3　X 

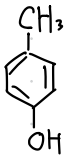
Y 

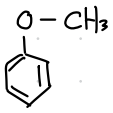
Z 

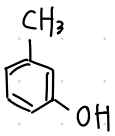
問4 

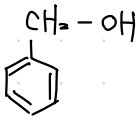
チ－13

問1　A： 

B： 

C： 

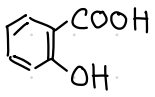
D： 

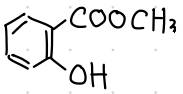
E： 

問2　NaOH aq を加えると A, B, D は中和して均一な溶液になる（とける）。
（C, E は2層に分離する）　↑これだけでは×。見た目や匂いで分かることを書く。

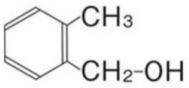
問3　ヒドロキシ基を持たないため，水素結合ができず，分子間力が弱いため。

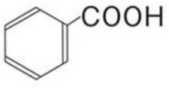
問4　金属 Na を加えたとき，C は H_2 を発生しない。

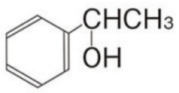
問5.　F：サリチル酸 

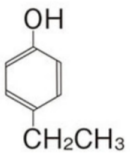
G：サリチル酸メチル 

チ－14

(1)　5　　(2) 

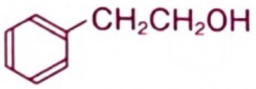
　　(3)　CHI_3　　(4) 

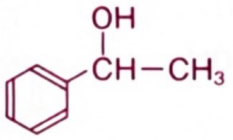
(5) 

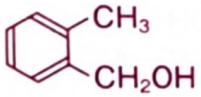
　　(6)　9　　(7) 

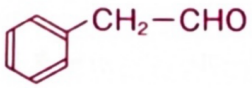
　　(8)　5　　(9)　1：2：2：2：3

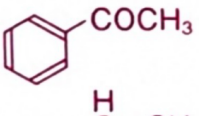
4－15

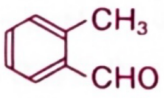
問1　A 

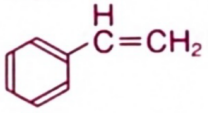
B 

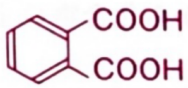
C 

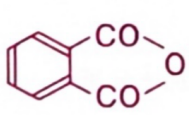
D 

E 

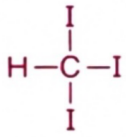
F 

G 

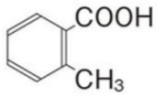
H 

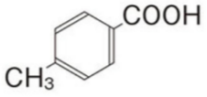
I 

問2　B　　**問3**　ポリスチレン

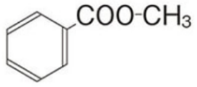
問4　記号　B　　構造式 

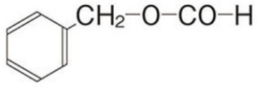
4－16

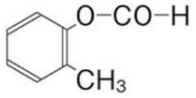
問1(1)　A 

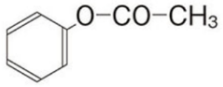
B 

(2)　ナフタレン，*o*－キシレン

問2(1)　C 

D 

E 

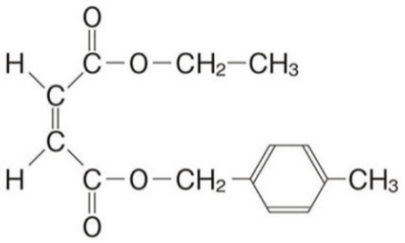
F 

(2)　臭素

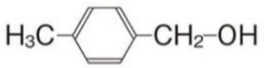
4 - 17

問1 $C_{14}H_{16}O_4$

問2 A

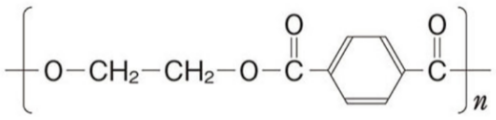


C



D H_3C-CH_2-OH

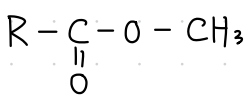
H



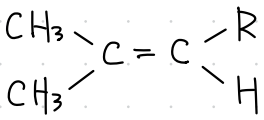
4 - 18

問1 $C_{12}H_{14}O_2$

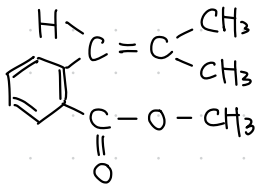
問2 (a)



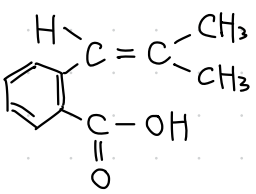
(b)



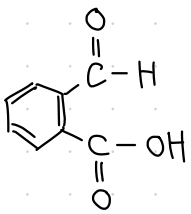
問3 A



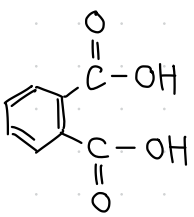
B



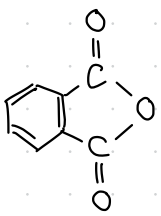
C



D



E



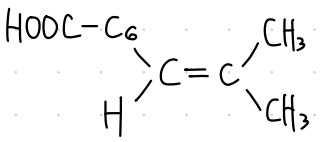
問1　$C_xH_yO_z$ とおくと,

$$x = \frac{190\,g \times \frac{75.8}{100}}{12\,g/mol} = 12,\quad y = \frac{190\,g \times \frac{7.4}{100}}{1.0\,g/mol} = 14,\quad z = \frac{190\,g \times \frac{16.8}{100}}{16\,g/mol} = 2$$

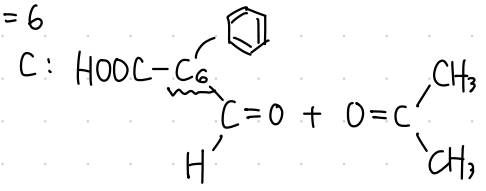
∴ $C_{12}H_{14}O_2$　$U = 6 =$  $\times 1 + C{=}C \times 1 + -\overset{\overset{O}{\|}}{C}-O-$

A $\xrightarrow{\text{加水分解}}$ { カルボン酸 B ($C_{11}H_{12}O_2$) ↑ A + H_2O − CH_4O ; メタノール }

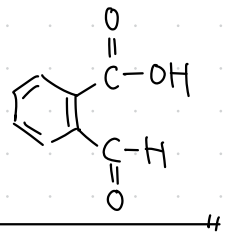
カルボン酸 B ($C_{11}H_{12}O_2$)　$U = 6$

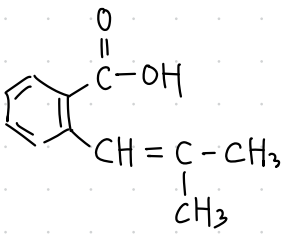


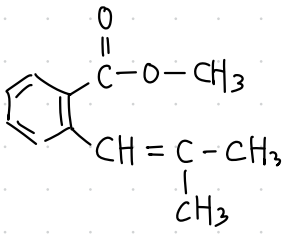
$\xrightarrow{O_3}$

C: 

D $\xrightarrow{-H_2O}$ E　∴ C はオルト体

C: 

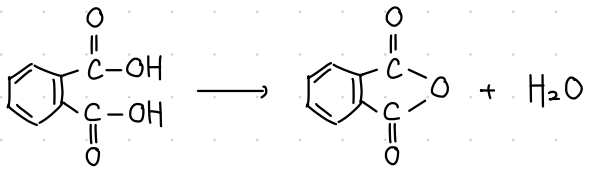
B: 

A: 

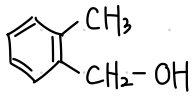
4 - 19

問1　あ　C_5H_6O　　い　164　　う　$C_{10}H_{12}O_2$

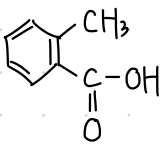
問2



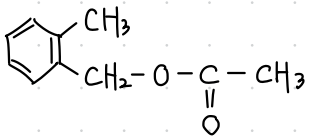
問3　C



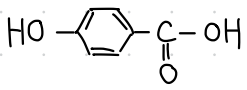
D



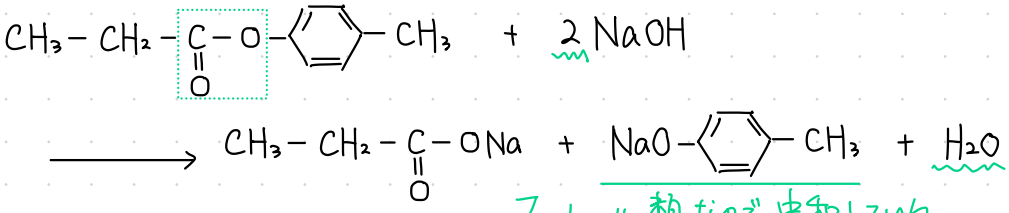
問4.



問5



問6



フェノール類なので中和している

問1　$\Delta t_f = k_f m$　　$2.56 = 5.12 \times \dfrac{\dfrac{410\times10^{-3}\,g}{M\ g/mol}}{5\times10^{-3}\,kg}$　　$\therefore M = 164$ //(い)

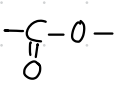
$C_xH_yO_z \xrightarrow{O_2} x\,CO_2 + \dfrac{y}{2}H_2O$ より

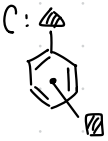
$C_xH_yO_z : CO_2 = 1 : x$ だから，　$\dfrac{8.2\,g}{164\ g/mol} \times x = \dfrac{22\,g}{44\ g/mol}$　　$\therefore x = 10$

$C_xH_yO_z : H_2O = 1 : \dfrac{y}{2}$ だから，　$\dfrac{8.2\,g}{164\ g/mol} \times \dfrac{y}{2} = \dfrac{5.4\,g}{18\ g/mol}$　　$\therefore y = 12$

$M = 12x + y + 16z = 120 + 12 + 16z = 164$　　$\therefore z = 2$

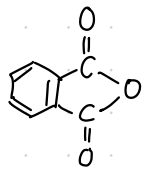
$\therefore$ $C_{10}H_{12}O_2$ //(う)　　C_5H_6O //(あ)

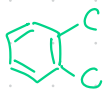
$U = 5 =$  $\times 4 +$ 

芳香族エステルA $\xrightarrow{けん化}$ ㊦ C： 

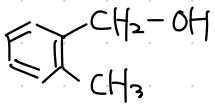
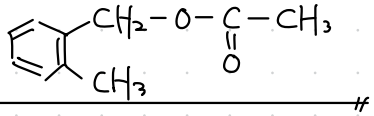
・アルコール（フェノール類）：$-C-OH$

・C（第1級）$\xrightarrow{Ox}$（弱く）酸性D（$C_8H_8O_2$）（カルボン酸）　C：C_8

・C $\xrightarrow{KMnO_4}$ E $\xrightarrow{-H_2O}$ F（$C_8H_4O_3$）：  （ホルト体）

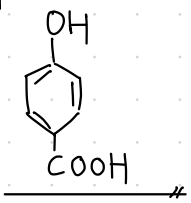


㊧ C′：カルボン酸のNa塩　$C_{10} - C_8 = C_2$

$\therefore$ C：  　C′：$CH_3-\overset{\overset{O}{\|}}{C}-OH$　$\therefore$ A： 

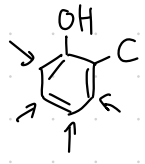
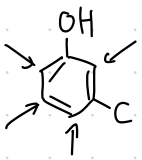
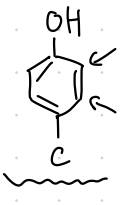
芳香族エステルB $\xrightarrow{けん化}$ ㊧ { フェノール類G のNa塩，カルボン酸G′ のNa塩 }

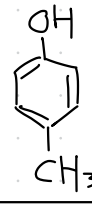
フェノール類のエステル　㊦がない

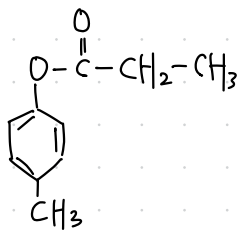
H： 

$\xrightarrow{H^+}$ { フェノール類 油状G（酸性，108）$\xrightarrow{KMnO_4}$ H（138）（カルボン酸・フェノール類） }

$-CH_3 \xrightarrow{+30} -COOH$

G：   

$\therefore$ G：（108） 

B： 

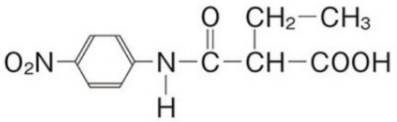
$G' = B + H_2O - G = C_{10}H_{12}O_2 + H_2O - C_7H_8O = C_3H_6O_2 =$ $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-OH$

4－20

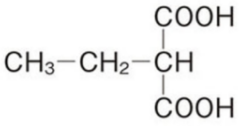
問1　C 原子…52.4 %　　H 原子…4.8 %　　N 原子…11.1 %

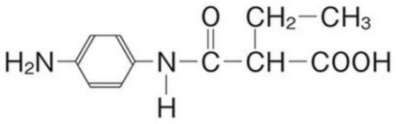
問2　$RCOOH + NaHCO_3 \longrightarrow RCOONa + CO_2 + H_2O$

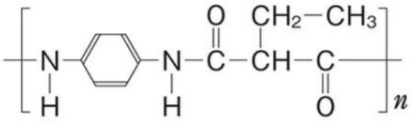
問3

A 

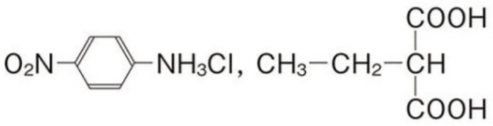
B 

C 

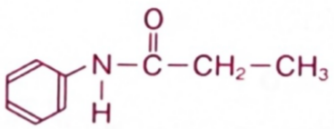
D 

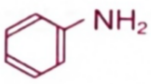
E 

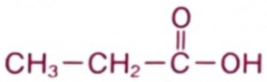
問4　塩酸を用いることで化合物 A のアミド結合が加水分解されてしまうため。

生成物　 

4－21

問1　A 

B 

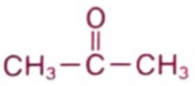
C 

D $CH_3-CH_2-CH_2-OH$

問2　 

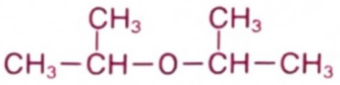
．31 mg

問3　F　$CH_3-CH_2-O-CH_3$，エチルメチルエーテル

G 

．アセトン

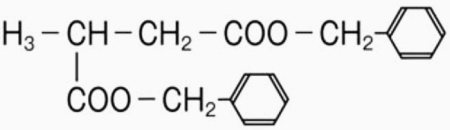
問4　F，E，D

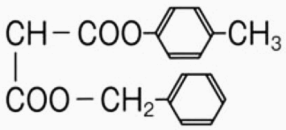
問5　H 

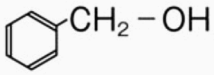
I　$CH_2=CH-CH_3$

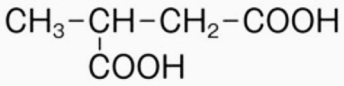
4－22

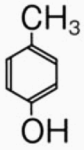
問1　$C_{19}H_{20}O_4$

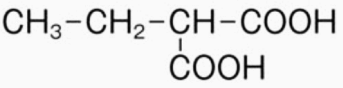
問2　A： 

B： 

C： 

D： 

E： 

F： 

G： 